SOLEWISE · FOOTWEAR INTELLIGENCE

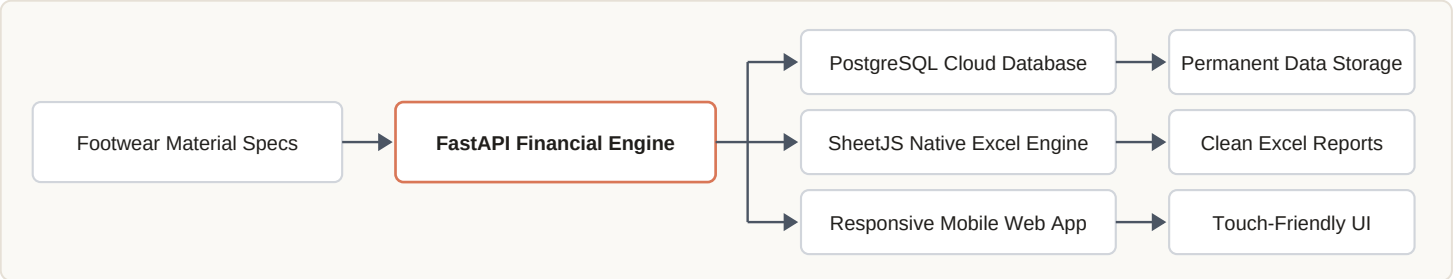
Project Overview, Problem Analysis & Technical Solutions

Status: Production Live
Platform: Web & Mobile Phone App
Live URL: sole-wise.onrender.com

Executive Summary: Solewise is a modern footwear cost intelligence platform designed to replace messy, manual spreadsheets with automatic material comparison and batch savings calculations. Footwear manufacturers can easily compare traditional leathers against modern synthetic alternatives, evaluate physical properties like thickness and flex cycles, and calculate exact production batch savings. This report details the key problems solved during development and how the application was made fast, reliable, and easy to use on both phones and computers.

60.7%	100%	13 / 13	33.1 kB
Direct Cost Savings (Synthetic vs Leather)	Permanent Cloud Sync (Render PostgreSQL)	Automated Tests Passing (Zero Fake Test Data)	Ultra-Fast Page Load (Smooth Mobile Experience)

1. System Data Flow & Architecture



2. Problems Identified in the System

• Disappearing Data on Free Cloud Server
When deployed on free cloud hosting, the server automatically sleeps after 15 minutes of inactivity. When it woke back up, it reset itself to the initial state, causing newly entered footwear materials and prices to vanish.

• Broken & Corrupted Excel Exports
Exported spreadsheets had broken formatting: rupee currency symbols turned into illegible garbled text, and column widths were too narrow, cutting off prices and numbers with errors.

• Test Data Polluting the Real Catalog
Automated background tests were accidentally saving dummy sample records directly into the live catalog, cluttering the screen with unwanted test entries that the user never added.

• Cluttered Layout & Poor Mobile Experience
The interface was packed with unnecessary menu items (Costing, Sequence, Suppliers) and wide desktop tables that cut off on phone screens with buttons too small to tap.

3. Solutions Built & Implemented

• Permanent Cloud Database (PostgreSQL)
Connected a dedicated cloud PostgreSQL database directly inside Render. Now, every footwear material, technical specification, and price entered by the user is permanently saved and never lost.

• Clean Native Excel (.xlsx) Downloads
Replaced broken text exports with real Microsoft Excel spreadsheets. Columns now automatically adjust to fit text, numbers are formatted cleanly, and prices display properly in Indian Rupees.

• Clean Database with Test Isolation
Separated the automated tests to run entirely in temporary test memory. Test sample shoes can never leak into the real catalog, keeping user data 100% clean and accurate.

• Modern Mobile-First Design & PWA
Redesigned the whole interface with warm, elegant typography, large touch-friendly buttons, a slide-out mobile menu, and an orange sneaker icon so it can be installed on phones like a native app.

4. Production Stack & Deployment Summary

Component	Technology	Role & Features
User Interface	React 18 + Tailwind CSS	Modern warm styling, responsive mobile drawer, clean Excel export, phone app support.
Backend Server	FastAPI (Python 3.14)	Fast web API, material alternative matching, automatic batch cost and yield calculations.
Database	PostgreSQL (Render Cloud)	Permanent cloud database, zero data loss, instant synchronization across phone and laptop.
Live Hosting	Render Cloud Web Service	Full-stack single domain hosting with automatic HTTPS security at sole-wise.onrender.com.